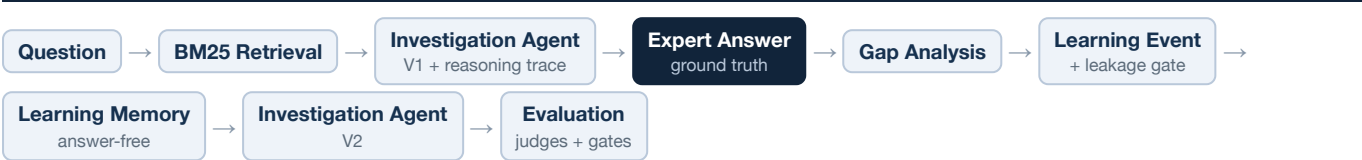


Problem. Organizational knowledge is scattered across Slack, tickets, wikis, and code. AI assistants retrieve facts but miss the causal reasoning an expert applies, so their answers disagree with experts and expertise stays trapped in individuals. FinFlow learns *how experts reason* from feedback and applies that method to new questions. The goal is learned reasoning, not harder search — and the proof is generalization to a held-out question the system never received feedback on.

Architecture



A plain linear pipeline, no orchestration framework. Key components:

- **LLM provider** — a pluggable interface; Groq (default `llama-3.3-70b-versatile`) plus a Mock provider for offline tests. Only the LLM package imports a vendor SDK.
- **Retrieval** — deterministic BM25 behind a `Retriever` interface: per-source balancing, soft routing, query expansion, persisted diagnosable snapshots. V2 adds an incident-only soft service-scope down-rank.
- **Learning** — distilled patterns split into retrieval signals and reasoning heuristics; a deterministic n-gram leakage gate drops any field overlapping the expert answer, so memory holds generalized method, never the answer.
- **Evaluation** — versioned LLM judges (similarity, root-cause rubric) plus deterministic evidence overlap, utilization, and V1-to-V2 deltas, reduced to four explicit gates.
- **Persistence and UI** — SQLite (9 tables); read-only and interactive Streamlit dashboards. 90 offline deterministic tests; one multi-mode Docker image.

Worked example (P2). "What caused the June 3 duplicate-charge incident?" V1 names the proximate cause (a Risk-Engine retry storm) but misses the disabled idempotency guard (commit a1), whose commit message has no lexical overlap with "duplicate charge". After the expert answer, the system learns the generic heuristic *check recent commits for removed safeguards*; V2 then retrieves commit a1 and names the true root cause and fix.

Assumptions

- Synthetic data only; the Slack feed is a simulated JSON file; human answers are ground truth; any reasonable LLM provider is acceptable; a UI is optional (provided anyway).
- BM25 (lexical) retrieval is chosen for determinism and reproducibility, with the interface left open for a future semantic retriever.
- The corpus is deliberately authored so each question has a piece of evidence reachable only via learned expansion, making the learning effect observable in a small dataset.
- LLM-as-judge similarity is subjective and noisy; scores are versioned and logged, and root-cause uses a fixed rubric. Live LLM outputs are not byte-reproducible.

Results (honest)

Class	Evidence
Verified (90 offline tests, deterministic)	Leakage-free generalizable learning; retrieval transfer (a held-out twin's V2 surfaces a commit V1 missed); evidence utilization; relevance gating; gate and persistence logic.
Directional (<code>llama-3.1-8b-instant</code> , P2 to H2, one family)	Held-out blended +0.325 , ablation +0.125 , rubric coverage 0.50 to 0.75 , similarity 0.40 to 0.80 , leakage PASS ; the newly-retrieved root-cause commit was used in V2 reasoning.
Not yet validated (canonical 70B, all families)	The full <code>llama-3.3-70b-versatile</code> run did not complete (Groq daily token quota). The only completed canonical run was pre-tuning and failed three of four gates.

Against the brief's success criteria (targets to aim for and reason about; partial achievement with analysis is the intent):

Metric	Target	FinFlow status (honest)
Similarity to expert	$\geq 75\%$	80% on H2 (8B, directional); not yet canonical
Root-cause identification	$\geq 70\%$	75% rubric coverage on H2 (8B, directional)
Evidence coverage	$\geq 80\%$	40% on H2 V2 (rose from 20%) — below target , stated openly
Learning improvement after feedback	$\geq 20\%$	+32.5 points blended on H2 (directional)
Response time	$< 15\text{ s}$	Single 8B investigations run in a few seconds; not formally benchmarked
Test-scenario coverage	100%	All 9 questions / 6 families exercised; 90 offline tests pass

Central claim: mechanism-verified and directionally supported, **not yet canonically validated** — stated plainly rather than overclaimed. No success is declared on the central claim until a canonical 70B run passes.

What I would build next

- Run the canonical 70B post-tuning evaluation once quota is available (the harness is ready).
- Expand the number of families and categories for statistical robustness.
- Add retry and backoff for transient API rate-limit errors.
- Add an optional semantic retriever behind the existing `Retriever` interface.
- Add a low-confidence / abstention path so the system fails gracefully instead of confabulating on out-of-distribution questions.
- Raise evidence coverage by sharpening retrieval-signal specificity toward the missed gold evidence.